The chain

01 Chip Design, EDA and IP
02 Silicon and Wafers
03 Chemicals, Gases and Photoresist
04 Lithography
05 Photomasks and Pellicles
06 Deposition and Etch
07 Metrology and Inspection
08 Transistors and the Front End
09 Foundries and Fabs
10 Memory and HBM
11 Advanced Packaging
12 Substrates and PCBs
13 Test and Assembly
14 Systems and Networking
15 Data Centers and Power

16 The Economics of AI Chips
17 Who Controls the Chain

Chapter 09 Stage 9 of 15 Extreme

Foundries and Fabs

TSMC's six largest plants are all in Taiwan. In 2025 they could make more than 13 million wafers a year, out of more than 17 million for the whole company.

Specimen 09 · Foundries and Fabs

A FOUP, the sealed pod that carries 25 wafers between tools, drawn to scale. The 300 mm wafers inside, each the width of a dinner plate, nearly fill it. The shell is translucent, as the real polycarbonate is. One wafer is picked out in silver.

In plain terms

A foundry is a company that makes chips for other companies in its chip factories, or fabs. Those companies send in their designs, and the foundry runs them all through the same building on the same tools. The building costs tens of billions of dollars, and the air inside is cleaner than in an operating room. Even so, each wafer, the silicon disc the chips are built on, moves from machine to machine inside sealed boxes and never touches that air. No chip company sells enough of one product to fill such a factory on its own, so chip companies share a handful of them. TSMC runs the biggest of these shared factories, and all six of its largest plants are in Taiwan. Taiwan also held most of the world's capacity for the most advanced logic chips in 2022. So a political crisis over Taiwan would put most of the supply of those chips at risk.

In short

In 2022 Taiwan held 69 percent of the world's capacity for advanced logic chips, those below 10 nanometers, and the United States almost none[14]. In July 2026 TSMC raised its planned investment in Arizona to $265 billion across 12 facilities[15].

The industry association SEMI projected in 2026 that China would keep the world's largest installed fab capacity in 2026 and 2027[16]. China's SMIC cannot buy an EUV scanner, the machine that prints the finest layers, yet it makes 7 nm-class chips by printing those layers in several passes on older deep-ultraviolet scanners[17].

14 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org
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Chokepoint card

Foundries and Fabs, stage 9 of 15

Concentration

Extreme

Three or fewer firms could supply this stage today, or one firm holds more than half of it.

Substitutability

Hard

TSMC has not built one of its six largest fabs outside Taiwan [1].

Price or market size

$35-43 billion

over ten years to own a new 2 nm-class fab completed in 2026, BCG's 2023 estimate [2]

Who leads	TW	**TSMC**	79% of its wafer revenue from 7 nm and below, 3Q26 [3]
	US	**Intel Foundry**	$5.8bn segment revenue in 2Q26, only $293m of it from outside customers [4]
	KR	**Samsung Foundry**	Over $37bn planned in Texas (CHIPS award, Dec 2024) [5]
	CN	**SMIC**	$9.327bn revenue in 2025 [6]
	JP	**Rapidus**	Raised 267.6 billion yen in February 2026 to reach 2 nm mass production by 2027 [7]
Where it is made	TW	**Taiwan**	TSMC (six gigafabs, N2 and N3 lines) and UMC
	KR	**South Korea**	Samsung Foundry, Hwaseong and Pyeongtaek
	CN	**China**	SMIC and Hua Hong
	US	**United States**	Intel Foundry, GlobalFoundries, TSMC Arizona, Samsung Taylor
	JP	**Japan**	TSMC Kumamoto, Rapidus Chitose

Why substitution is slow

In March 2026 the Semiconductor Industry Association (SIA) put the cost to build and equip a leading-edge fab at as much as $20 to $25 billion [8]. SemiAnalysis judged in December 2025 that finding the highly skilled engineers a foundry needs remained a major challenge for TSMC in Arizona [9].

Where China stands

Georgetown University's CSET reported in June 2024 that Huawei's Ascend 910B AI chip is believed to be made on SMIC's 7 nm-class N+2 process [10]. But SemiAnalysis judged in April 2025 that most 910B and 910C chips were made at TSMC [11].

Where the US stands

The US government agreed in August 2025 to buy a passive 9.9 percent stake in Intel, with no board seat [12]. Announcing the deal, Intel called itself the only chipmaker that both develops and manufactures leading-edge logic chips in the United States [12]. Intel Foundry lost $2.089 billion in the second quarter of 2026 [13].

1 Fab Capacity. TSMC, accessed 9 September 2026. tsmc.com

2 Navigating the Costly Economics of Chip Making. Boston Consulting Group, 28 September 2023. bcg.com

3 TSMC Reports Third Quarter EPS of NT$29.43. U.S. Securities and Exchange Commission, 15 October 2026. sec.gov

4 INTEL CORP, Form 10-Q quarterly report for the period ended 2026-06-27 (10-Q). U.S. Securities and Exchange Commission, 24 July 2026. sec.gov

5 Biden-Harris Administration Announces CHIPS Incentives Award with Samsung... NIST, 20 December 2024. nist.gov

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7 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private... Rapidus, 27 February 2026. rapidus.inc

8 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 3. epw.senate.gov

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10 Pushing the Limits: Huawei's AI Chip Tests U.S. Export Controls. Center for Security and Emerging Technology, 17 June 2024. cset.georgetown.edu

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12 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com

13 Intel Reports Second-Quarter 2026 Financial Results. Intel, 23 July 2026. intc.com

TSMC calls a plant a GIGAFAB once it can run more than 100,000 wafers a month, each a 300 mm disc, twelve inches across[18]. It operates six of them, and every one is in Taiwan[1].

01

How it works

Inside, a chip is built up in layers, and a similar loop of steps repeats for each one. Wafers travel in batches, called lots, inside a sealed pod called a FOUP, short for front-opening unified pod. Entegris, a US maker of these pods, offers a standard version that holds 25 wafers[19]. An overhead track carries the pod from tool to tool.

At each tool a robot lifts one wafer out and the tool does its one job, such as depositing a film, printing a pattern, etching it, cleaning it or measuring it. Then the robot puts the wafer back. The pod moves on to the next tool, often across the building. When a layer is finished, the loop starts again with the next one.

Turning silicon into a finished, packaged chip takes 1,000 to 1,500 steps, depending on how complex the chip is, the Semiconductor Industry Association said in 2025[20]. In 2021 the association put the time to make a finished wafer at about 12 weeks on average, and up to 14 to 20 weeks for advanced processes[21].

The pod keeps the wafer clean for all of that time because the wafer never meets the room's air. The room around the pods is kept clean as well. Intel said in 2018 that the air in its fab cleanrooms is filtered to 1,000 times fewer airborne particles than a sterile hospital operating room[22].

Each tool is set by a recipe, the list of gases, temperatures and times for one step. A foundry offers every customer the same list of recipes, so every design in the building can run on the same shared tools. The tools run day and night, because an idle tool earns nothing. The finished wafer is tested (see Test and assembly), then cut and packaged.

A new leading-edge fab costs tens of billions of dollars. In 2023 the Boston Consulting Group (BCG) estimated that a new fab for chips at 2 nm and below, completed in 2026, would cost $35 to $43 billion to build, equip and run over ten years[2]. BCG said that was 33 to 66 percent more than the same cost in 2023[2]. Building and equipping alone takes as much as $20 to $25 billion of capital for a leading-edge fab, the Semiconductor Industry Association (SIA) told the US Senate in March 2026[8].

1 Fab Capacity. TSMC, accessed 9 September 2026. tsmc.com

2 Navigating the Costly Economics of Chip Making. Boston Consulting Group, 28 September 2023. bcg.com

8 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 3. epw.senate.gov

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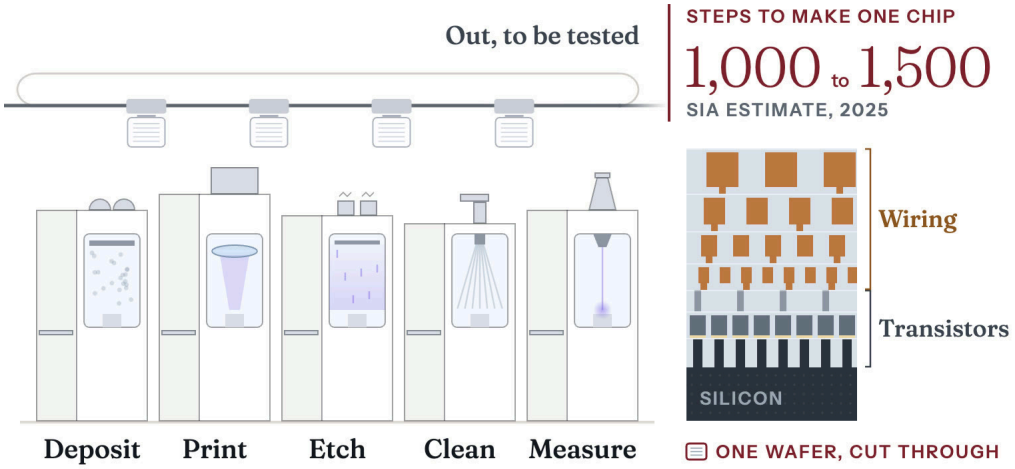
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21 Chipmakers Are Ramping Up Production to Address Semiconductor Shortage. Here's... Semiconductor Industry Association, 26 February 2021. semiconductors.org

22 Inside An Intel Chip Fab: One Of The Cleanest Conference Rooms On Earth. Intel, 28 March 2018. intc.com

Figure 09.1 · Explainer, 20 seconds

How a fab builds a chip, layer by layer



1. **The fab.** Sealed pods carry wafers from tool to tool on a track near the ceiling.
2. **Deposit.** The first tool lays a thin film on the wafer.
3. **Print.** The next prints a pattern on it with light.
4. **Etch.** An etch tool cuts the pattern into the film.
5. **Clean, measure.** The wafer is cleaned, then measured to check the layer came out right.
6. **Next layer.** Then the loop starts again for the next layer. The transistors come first, then the wiring above them.
7. **Finished.** Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025.

Note Simplified. A real fab has many tools of each kind, spread across the building, and a chip has far more layers than the seven shown. The layers are drawn far thicker than they are.

Most of TSMC's capacity is in its six largest plants, all of them in Taiwan. Together, the plants TSMC and its subsidiaries run had capacity for more than 17 million wafers a year in 2025, counted as if each were the standard 12-inch size[1]. The six largest plants had a combined capacity of more than 13 million[18]. The other plants are five smaller fabs in Taiwan, two in China, two in the United States and one in Japan[1].

02

Variants and trade-offs

Pure-play foundries

TSMC and UMC, both based in Taiwan, make chips only for other companies. TSMC, founded in 1987, pioneered this pure-play model[23]. Chips at 7 nm and below, meaning the newest processes, were 79 percent of TSMC's wafer

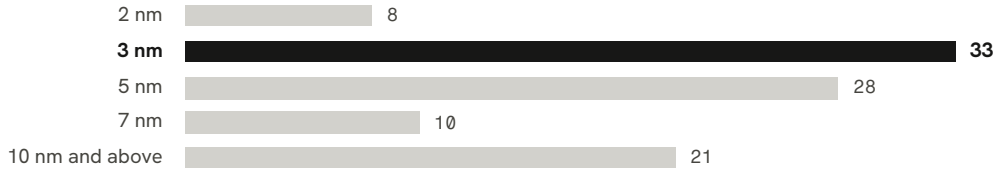
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revenue in the third quarter of 2026[3]. That share split into 33 percent at 3 nm, 28 at 5 nm, 10 at 7 nm and 8 at 2 nm[3]. TSMC's revenue for the quarter was $46.72 billion, at a 66.4 percent gross margin[3].

Figure 09.2

3 nm led TSMC's wafer revenue in third-quarter 2026, at 33 percent

TSMC wafer revenue by node, 3Q26 % of wafer revenue



Source TSMC 3Q26 results, filed with the SEC [3]

Integrated device makers turned foundry

Intel and Samsung design and sell their own chips and rent out capacity too. Renting out capacity is harder for them, for two reasons. Customers dislike handing their designs to a competitor, and each company's own products take its best capacity. Intel Foundry reported $5.8 billion of segment revenue in the second quarter of 2026 and an operating loss of $2.089 billion[13]. Only $293 million of that revenue came from outside customers[4]. Intel says substantially all of its foundry business still makes Intel's own products[4].

National champions

China's SMIC and Hua Hong and Japan's Rapidus have government backing. SMIC cannot buy an extreme ultraviolet (EUV) scanner, the machine that prints the finest layers[17]. Yet it makes 7 nm-class chips on its N+2 and N+3 processes by printing those layers in several passes on older deep-ultraviolet scanners[17]. SMIC's revenue rose 16.2 percent to $9.327 billion in 2025, at a 21 percent gross margin[6]. Its lines ran 93.5 percent full, on monthly capacity above a million wafers counted in the smaller 8-inch size[6].

Two research groups disagree about where Huawei's Ascend AI chips are made. Georgetown University's Center for Security and Emerging Technology (CSET) reported in June 2024 that Huawei's Ascend 910B AI chip was believed to be made on SMIC's N+2 process[10]. It also reported that SMIC's limited 7 nm capacity forced Huawei to choose between its most advanced phone chips and its AI chips[10]. But the research firm SemiAnalysis judged in April 2025 that most Ascend 910B and 910C chips were made on TSMC's 7 nm process[11]. It said Huawei got around the sanctions that barred it from TSMC by buying about $500 million of 7 nm wafers through another company, Sophgo[11].

3 TSMC Reports Third Quarter EPS of NT$29.43. U.S. Securities and Exchange Commission, 15 October 2026. sec.gov

4 INTEL CORP, Form 10-Q quarterly report for the period ended 2026-06-27 (10-Q). U.S. Securities and Exchange Commission, 24 July 2026. sec.gov

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11 Huawei AI CloudMatrix 384. SemiAnalysis, 16 April 2025. newsletter.semianalysis.com

13 Intel Reports Second-Quarter 2026 Financial Results. Intel, 23 July 2026. intc.com

17 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

In February 2026 Rapidus raised 267.6 billion yen from the Japanese government and 32 companies to reach 2 nm mass production by 2027[7].

Mature-node capacity

Most wafers come off older lines. These lines make power devices, analog chips and microcontrollers, and the foundries that run them compete on price.

The industry association SEMI had expected part of the growth in foundry capacity to come from Intel, which later scaled back. In January 2025 SEMI forecast that foundry capacity at all nodes would rise from 11.3 million wafers a month in 2024 to a record 12.6 million in 2025, counted in the 8-inch size[24]. SEMI's earlier forecast, from June 2024, had credited much of the expected growth to Intel's new foundry business and China's expansion[25]. But in July 2025 Intel scaled back its own expansion plans, saying it would no longer go ahead with planned projects in Germany and Poland and would slow construction in Ohio[26].

03

Capacity and cost

In March 2025 TSMC announced plans to add $100 billion to the $65 billion it was already investing in Phoenix, for three more fabs, two packaging plants and an R&D center[27]. In July 2026 it announced another $100 billion for four more fabs at 2 nm or below, taking its planned investment in Arizona to $265 billion across 12 facilities[15].

Some customers pay TSMC in advance to hold capacity at its fabs[28]. These deposits stood at NT$189,858.2 million at the end of 2025, down from NT$291,101.8 million a year earlier[28].

In 2025 the biggest AI chip designers used most of the world's supply of two inputs but only a small share of its advanced chipmaking. Epoch AI estimated that the four largest used around 90 percent of the world's CoWoS packaging capacity and HBM memory supply[29]. CoWoS is TSMC's packaging for AI chips, and HBM is the stacked memory that sits beside them. The same four used only 12 percent of advanced logic production[29]. See Advanced packaging.

7 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private... Rapidus, 27 February 2026. rapidus.inc

15 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

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28 TAIWAN SEMICONDUCTOR MANUFACTURING CO LTD, Form 20-F annual report for the... U.S. Securities and Exchange Commission, 16 April 2026. sec.gov

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04

Who makes it

In August 2025 the US government agreed to buy 433.3 million newly issued Intel shares at $20.47 each, or $8.9 billion in all, for a 9.9 percent stake[12]. It would pay with $5.7 billion of CHIPS Act grants that Intel had not yet received and $3.2 billion awarded to Intel under the Secure Enclave program[12]. Intel said it would keep meeting its Secure Enclave obligations and delivering trusted and secure chips to the US Department of Defense[12]. Announcing the deal, Intel's chief executive, Lip-Bu Tan, called Intel the only semiconductor company that both develops and manufactures leading-edge logic chips in the United States[12]. The deal closed on 27 August 2025 and made the government one of Intel's largest stockholders, Intel said in its annual report for 2025[30].

The stake is passive, with no board seat[12]. The government agreed to vote with Intel's board on matters that need shareholder approval, with limited exceptions[12]. The deal also gave the government a five-year warrant, a right to buy a further 5 percent of Intel at $20 a share[12]. The government can use that right only if Intel's ownership of its foundry business falls below 51 percent[12].

In 2026 Intel began high-volume manufacturing of some of its Panther Lake processors using ASML's High-NA extreme ultraviolet scanners[13]. It also put its 18A-P process into risk production, the trial runs a process goes through before it is signed off[13].

In 2022 Samsung broke ground in Taylor, Texas, on a plant with an initial minimum investment of $17 billion, split into $6 billion for buildings and $11 billion for equipment[31]. In December 2024 the US Commerce Department awarded Samsung up to $4.745 billion of CHIPS Act funding, to support more than $37 billion of planned investment in Central Texas[5]. That plan covers two new leading-edge logic fabs and a research fab in Taylor, and an expansion of Samsung's existing plant in Austin[5].

05

The chokepoint

In 2022 South Korea and Taiwan held almost all of the world's logic capacity below 10 nm[32]. The United States had almost no capacity below 10 nm that year[32]. In May 2024 SIA and BCG projected that the US share of that capacity

5 Biden-Harris Administration Announces CHIPS Incentives Award with Samsung... NIST, 20 December 2024. nist.gov

12 Intel and Trump Administration Reach Historic Agreement to Accelerate American... Intel, 22 August 2025. intc.com

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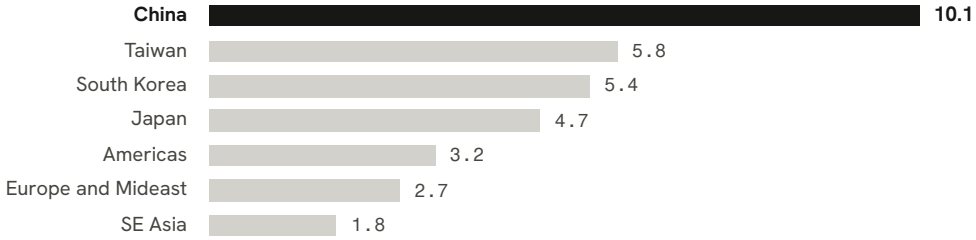
would reach 28 percent by 2032[14]. They also projected that new fab investment worldwide would reach around $2.3 trillion from 2024 to 2032[33].

China leads the world in capacity for logic chips made at 28 nm or older, but little of its capacity is advanced[34]. Across all nodes, old and new, SEMI forecast in June 2024 that China would have installed capacity of 10.1 million wafers a month in 2025, the most of any region[25]. It forecast 5.8 million for Taiwan, out of a world total of 33.7 million, all counted in the 8-inch size[25]. In the third-quarter 2026 update of its World Fab Forecast, SEMI projected that China would keep the largest installed capacity in 2026 and 2027[16].

Figure 09.3

China led SEMI's 2025 capacity forecast at 10.1 million 8-inch-equivalent wafers monthly

SEMI's June 2024 forecast of installed wafer fab capacity by region in 2025, all nodes million 8-inch-equivalent wafers per month



Source SEMI World Fab Forecast, June 2024 [25]

Chip production in Taiwan faces three risks:

- **Seismic.** Taiwan sits on an active plate boundary, and after an earthquake in January 2025 TSMC had to scrap some wafers in process[35].
- **Power.** TSMC says power outages, dips or surges caused by problems at its electricity supplier have at times interrupted its operations[28]. It warns that power outages or shortages could also limit its expansion[28].
- **Political.** Taiwan held 69 percent of the world's logic capacity below 10 nm in 2022[14]. In December 2025 SemiAnalysis put Taiwan's share of the world's advanced-node capacity at over 90 percent, though it did not say which node sizes count as advanced[9]. So a blockade or conflict would cut most of that output off from the world market at once. See Who Controls the Chain.

Building capacity outside Taiwan is slow, and TSMC has yet to build a gigafab anywhere else[1]. SemiAnalysis wrote in December 2025 that a foundry needs

1 Fab Capacity. TSMC, accessed 9 September 2026. tsmc.com
9 TSMC Overseas Fabs – A Success? SemiAnalysis, 1 December 2025. newsletter.semianalysis.com
14 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org
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35 TSMC January 2025 Revenue Report and the Statement on the Impact of Earthquake. TSMC, 10 February 2025. pr.tsmc.com

highly skilled engineers to get the most working chips from each wafer[9]. It judged that finding enough workers remained a major challenge for TSMC in Arizona[9].

Four questions to answer from memory

1. What is a foundry? see How it works
2. Where are TSMC's largest plants? see How it works
3. What is a major challenge for TSMC's plants in Arizona? see The chokepoint
4. How much of China's chipmaking capacity is advanced? see The chokepoint

Four things to remember

- A foundry makes chips for other companies, because no chip company sells enough of one product to fill a fab alone.
- All six of TSMC's largest plants are in Taiwan.
- SemiAnalysis judged in December 2025 that finding enough skilled workers remained a major challenge for TSMC in Arizona.
- SEMI projected in 2026 that China would keep the world's largest installed fab capacity in 2026 and 2027, but little of it is advanced.

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A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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